

CPS 200

Lecture 02

Jan 14, 2026

Kyu Hur

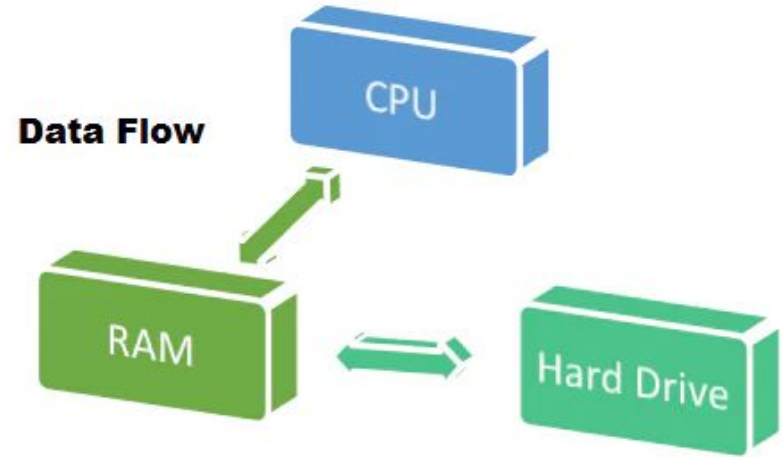
Schedule

Date	Session	Quiz	Topic
Jan 5			Reading Day
Jan 7	1	No	Intro to Programming I
Jan 12			National Holiday
Jan 14	2	No	Think Python: Programming as a way of thinking, Variables and Statements
Jan 19	3	Yes	Think Python: Functions, Functions and Interfaces
Jan 21	4	Yes	Think Python: Conditionals and Recursion, Return Values
Jan 26	5	Yes	Think Python: Iteration and Search, Strings and Regular Expressions
Jan 28	6	Yes	Think Python: Lists, Dictionaries
Feb 2	7	Yes	Think Python: Tuples, Text Analysis and Generation
Feb 4	8	Yes	Think Python: Files and Databases, Classes and Functions

From Lecture 01

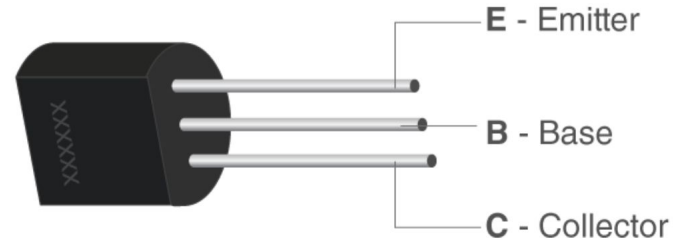
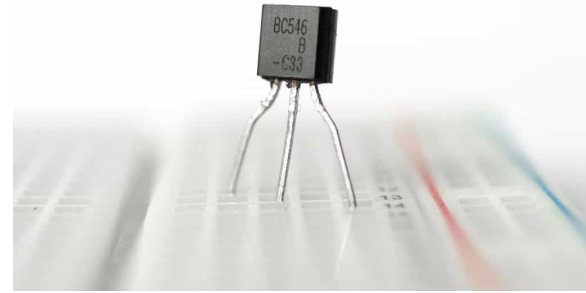
Parts of a Computer

- A computer is built of multiple components, but you can think of it as three fundamental parts:
 - **Central Processing Unit (CPU)**: the “brain”
 - **Random Access Memory (RAM)**: short-term memory
 - **Storage (SSD/HDD)**: long-term memory



Transistors

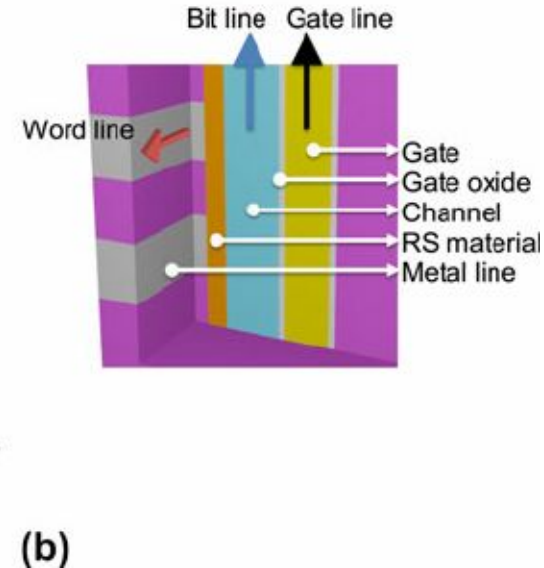
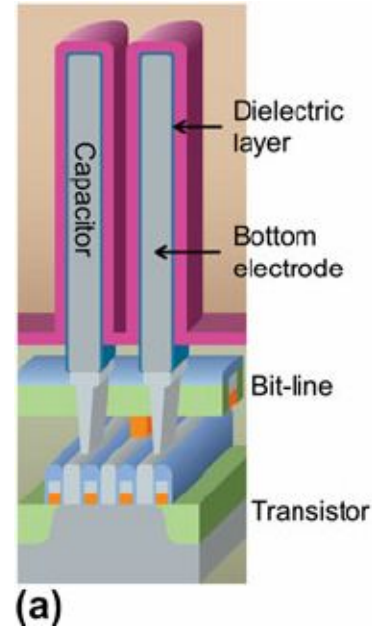
- A **transistor** is a tiny electronic switch; which allows electricity to flow through or not
- **Transistors** are the base of all computers today
- More transistors = “stronger” computation



Transistors in Data Storage

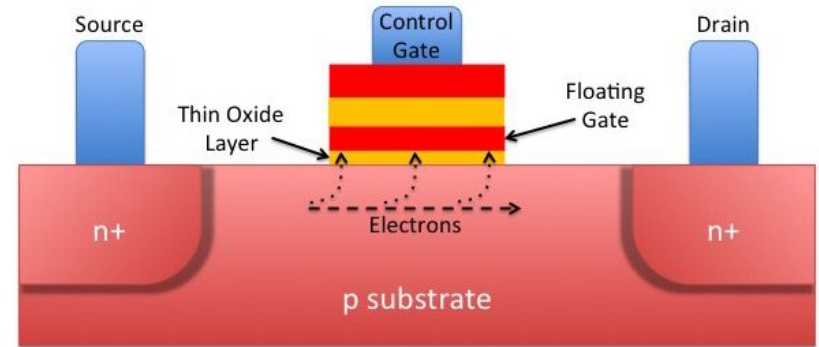
Transistors in RAM

- **Transistors** are used to process and store information
- **Capacitors** hold the charge but leak the charge naturally over time
- Systems refresh every ~64 ms (i.e. if power is cut, data is lost)



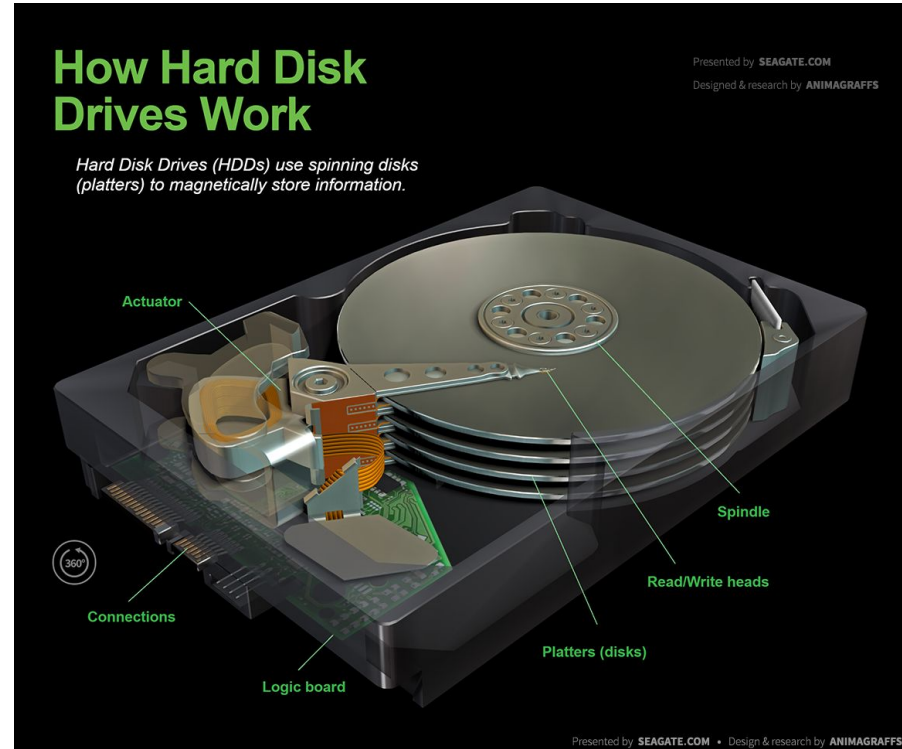
Transistors in SSD (Long-term storage)

- Floating-gate **transistors** has a control and floating gate
- Electrons are physically trapped in the floating gate
- Trapped charge changes the voltage of transistors
 - No charge $\rightarrow 0$
 - Charge $\rightarrow 1$



Transistors in HDD (Long-term storage)

- Data is stored on magnetic discs
- Data is represented in magnetic domains ($\sim 50 \text{ nm}^2$)
- Discs spin at 5400-7200 rpm, with the actuator working in tandem to read the correct sections of the disc



What is Binary?

What is Binary?

- What is **binary**? First, try explaining (in your own words) what binary is to your partners.
 - Semantic meaning of the word
 - Meaning in computer science
- What may be some other ways to represent numbers? For example, **ternary**? **Quaternary**?

What is Binary?

- **Binary** is a representation of information through a number system with only two possible values.
 - **Binary:** 0, 1
 - **Ternary:** 0, 1, 2
 - **Quaternary:** 0, 1, 2, 3
 - **Quinary:** 0, 1, 2, 3, 4
 - etc.
- What do we call a system that represents 10 possible values? (i.e. 0, 1, 2, 3, 4, 5, 6, 7, 8, 9)?

What is Binary?

- We call it a **decimal** system
- The ASCII table from last week had a “**decimal**” column

Intro to Programming I – What is a Computer?

ASCII TABLE

Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char
0	0	[NULL]	32	20	[SPACE]	64	40	@	96	60	`
1	1	[START OF HEADING]	33	21	!	65	41	A	97	61	a
2	2	[START OF TEXT]	34	22	"	66	42	B	98	62	b
3	3	[END OF TEXT]	35	23	#	67	43	C	99	63	c
4	4	[END OF TRANSMISSION]	36	24	\$	68	44	D	100	64	d
5	5	[ENQUIRY]	37	25	%	69	45	E	101	65	e
6	6	[ACKNOWLEDGE]	38	26	&	70	46	F	102	66	f
7	7	[BELL]	39	27	'	71	47	G	103	67	g
8	8	[BACKSPACE]	40	28	(	72	48	H	104	68	h
9	9	[HORIZONTAL TAB]	41	29	)	73	49	I	105	69	i
10	A	[LINE FEED]	42	2A	*	74	4A	J	106	6A	j
11	B	[VERTICAL TAB]	43	2B	+	75	4B	K	107	6B	k
12	C	[FORM FEED]	44	2C	,	76	4C	L	108	6C	l
13	D	[CARRIAGE RETURN]	45	2D	-	77	4D	M	109	6D	m
14	E	[SHIFT OUT]	46	2E	.	78	4E	N	110	6E	n
15	F	[SHIFT IN]	47	2F	/	79	4F	O	111	6F	o
16	10	[DATA LINK ESCAPE]	48	30	0	80	50	P	112	70	p
17	11	[DEVICE CONTROL 1]	49	31	1	81	51	Q	113	71	q
18	12	[DEVICE CONTROL 2]	50	32	2	82	52	R	114	72	r
19	13	[DEVICE CONTROL 3]	51	33	3	83	53	S	115	73	s
20	14	[DEVICE CONTROL 4]	52	34	4	84	54	T	116	74	t
21	15	[NEGATIVE ACKNOWLEDGE]	53	35	5	85	55	U	117	75	u
22	16	[SYNCHRONOUS IDLE]	54	36	6	86	56	V	118	76	v
23	17	[END OF TRANS. BLOCK]	55	37	7	87	57	W	119	77	w
24	18	[CANCEL]	56	38	8	88	58	X	120	78	x
25	19	[END OF MEDIUM]	57	39	9	89	59	Y	121	79	y
26	1A	[SUBSTITUTE]	58	3A	:	90	5A	Z	122	7A	z
27	1B	[ESCAPE]	59	3B	;	91	5B	[	123	7B	{
28	1C	[FILE SEPARATOR]	60	3C	<	92	5C	\	124	7C	
29	1D	[GROUP SEPARATOR]	61	3D	=	93	5D	]	125	7D	}
30	1E	[RECORD SEPARATOR]	62	3E	>	94	5E	^	126	7E	~
31	1F	[UNIT SEPARATOR]	63	3F	?	95	5F	_	127	7F	[DEL]

What is Binary?

- **Decimal** system is arbitrary
 - Likely due to humans having ten fingers, so it naturally evolved in that manner
- A better numbering system could have been a **Duodecimal** (12 states) or **Senary** (6 states) system
 - Because these numbers divisible by 1, 2, 3, 6, and 12 (for duodecimal)
 - Decimal system is only divisible by 1, 5, 10

What is Binary?

- Logarithms are usually represented by base 10, e.g.
 - **Log₁₀**
- Logarithms are also represented by base 2, e.g.
 - **Log₂**, also known as **ln** or the **natural logarithm**
- Logarithm bases are simply representations of these numbering systems (the base = type of numbering system)

Think Python

Google Colab Setup

- Please open up your computers and launch **Google Colab**
- [Introduction to *Think Python* on Jupyter](#)
- [Chapter 01.ipynb](#)

Think Python: Programming as a Way of Thinking

- [Chapter 01.ipynb](#)

Think Python: Variables and Statements

- [Chapter 02.ipynb](#)

End